Introducing marimo

A Next-Generation Reactive Notebook for Python

Michael Powers | Cville Data Science Meetup

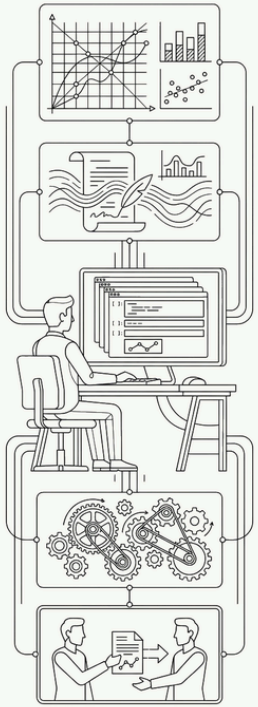
Why Do We Use Notebooks?

Explore data interactively – write code, see results inline, iterate without context-switching

Tell a story with data – interleave narrative, code, and visuals into a single document

Prototype and experiment – try ideas rapidly; the notebook is the scratchpad and the deliverable

Share reproducible work – hand someone a notebook, they re-run your analysis and build on it



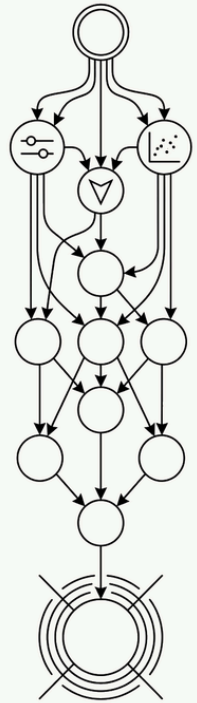
How marimo Makes Notebooks Better

Explore – Reactive execution: change a widget and every dependent cell updates automatically. First-class sliders, dropdowns, and selectable plots.

Prototype – Iterate by changing inputs and watching results propagate. No manual re-running in the right order.

Tell a story – Pure `.py` files. App mode hides code; slides layout turns cells into a slide deck. Deploy as a web app.

Share reproducibly – Self-contained environment via **PEP 723** + **uv**. One file, no setup instructions.

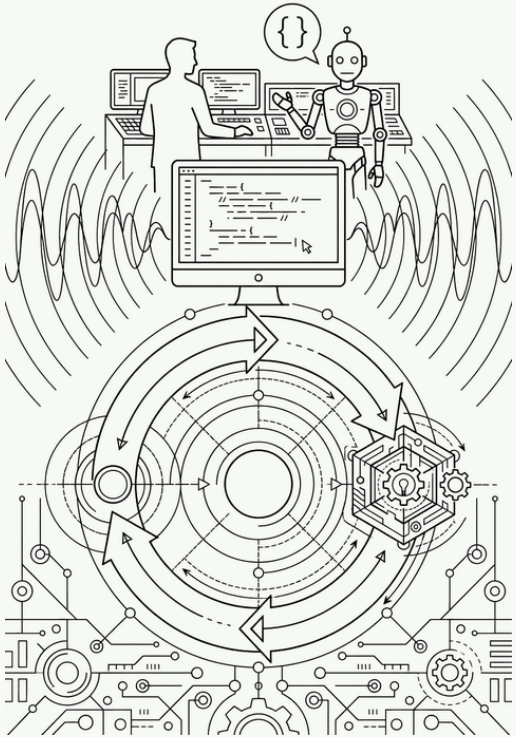


AI-Native by Design

`.py` format is why this works – AI agents generate and edit **plain Python**, not fragile JSON

Built-in AI assistant has access to live runtime variables, not just static code

Claude Code + `CLAUDE.md` prompt + `--watch` mode = describe what you want, see it render live



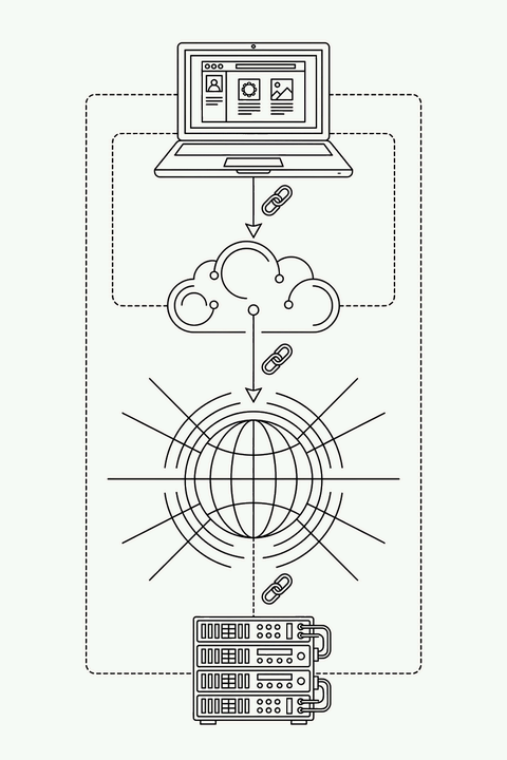
Deploy and Share

`marimo run notebook.py` → serve as a **web app**

molab (molab.marimo.io) → free cloud hosting, share by link, preview from GitHub

WASM export → static HTML on GitHub Pages, no backend needed

Cloudflare runs marimo at **notebooks.cloudflare.com**



Live Demo

Three pieces, one narrative: **from zero to interactive data app**

Build from scratch with AI

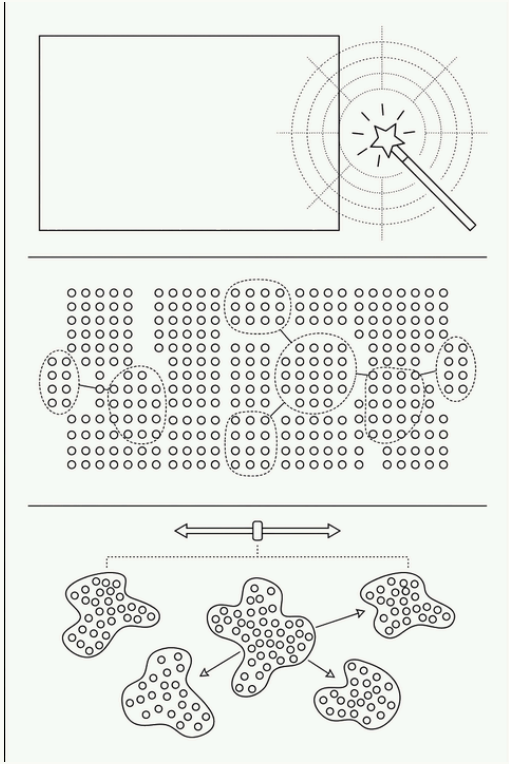
- Start with an empty file, use Claude Code `--watch` and marimo's AI assistant

MNIST Embedding Explorer

- Gallery notebook on molab – explore digit embeddings, 2D projection, cluster by digit class

Interactive K-Means

- Pre-built notebook with sliders to adjust K – watch clusters recolor live



Get Started

Install: `uv tool install marimo`

Create: `marimo edit --watch notebook.py`

Run as app: `marimo run notebook.py`

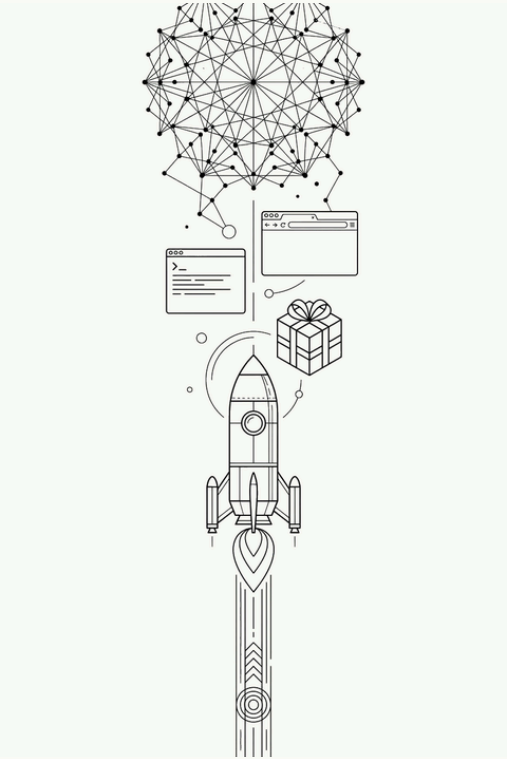
Sandbox: `marimo edit --sandbox notebook.py`

Docs: docs.marimo.io

Gallery: marimo.io/gallery

YouTube: youtube.com/@marimo-team

This demo: github.com/roadfoodr/marimo-demo



Thank You

Michael Powers

mpowers.info | github.com/roadfoodr/marimo-demo